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Neurobiological Correlates of Breathing Practices on Cognitive Functions: A Narrative Review

Dwivedi Krishna*, Saurav Nilesh & Kanthi Amit

Department of Yoga and Life Science, Swami Vivekananda Yoga Anusandhana Samsthana (S-VYASA), Bangalore, India

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Abstract- Breathing practices are rooted in ancient traditions such as yoga, tai chi, and qigong, and are increasingly recognized for their potential to enhance cognitive functions through neurobiological mechanisms. It influences the Autonomic Nervous System (ANS) by modulating Heart Rate Variability (HRV) and Respiratory Sinus Arrhythmia (RSA), both markers of autonomic balance and stress resilience. Enhanced vagal tone is associated with improved HRV and RSA which correlates with better attention, emotion regulation, and executive functions. Neuroplasticity and neurogenesis are also influenced by breathing practices. These processes are supported by increased levels of Brain-Derived Neurotrophic Factor (BDNF), which are linked to enhanced cognitive functions. Functional neuroimaging studies, such as fMRI and EEG, reveal that controlled breathing activates brain regions crucial for cognitive control, emotion regulation, and attention, including the prefrontal cortex and hippocampus. Clinical studies support the cognitive benefits of regular breathing practices, showing improvements in attention, memory, and executive function across diverse populations, including healthy individuals and those with cognitive impairments. The findings underscore the potential of breathing practices as non-pharmacological interventions to enhance cognitive functions and mitigate age-related cognitive decline. This review bridges the gap between ancient practices and modern neuroscience, offering evidence-based insights into the cognitive benefits of conscious breathing and its applications in clinical settings.

Key words: Neurobiology, Breathing Practices, Cognitive Functions

INTRODUCTION

Breathing practices have been integral to various traditional and modern health practices. Historically, these techniques are central to ancient disciplines such as yoga (pranayama), tai chi, and qigong, each emphasizing the regulation of breath to achieve physical, mental, and spiritual benefits. While the primary function of breathing is to ensure adequate oxygen supply and carbon dioxide removal, controlled breathing practices extend beyond mere respiratory functions.[1] They engage the Autonomic Nervous System (ANS) and have profound effects on the Central Nervous System (CNS), influencing cognitive functions and overall well-being. In recent years, scientific interest in breathing practices has surged, driven by growing evidence of their potential therapeutic effects on physical and mental health, particularly cognitive functions.[2]

Cognitive functions encompass a range of mental processes including attention, memory, executive functions, and problem-solving skills. These functions are crucial for daily life, academic performance, professional success, and overall quality of life. Cognitive decline,

*Corresponding author :
Phone : 9490253179
E-mail : krishnadwivedi@svyasa.edu.in

whether due to aging, neurodegenerative diseases, or other factors, poses significant challenges to individuals and society. Therefore, understanding and enhancing cognitive functions through non-pharmacological means, such as breathing practices, holds considerable promise.

Breathing is a unique physiological function that is both autonomic and voluntarily controlled. The respiratory centers in the brainstem, primarily the medulla oblongata and the pons, regulate the involuntary aspects of breathing.[3] Voluntary control of breathing is mediated by the motor cortex. This dual control allows breathing practices to modulate the ANS, which in turn influences Heart Rate Variability (HRV), a marker of autonomic balance and stress resilience. The interplay between the respiratory and central nervous systems provides a mechanistic basis for the potential cognitive benefits of breathing practices. Respiratory Sinus Arrhythmia (RSA) refers to the natural increase in heart rate during inhalation and decrease during exhalation.[4] RSA is a non-invasive marker of vagal tone, reflecting Parasympathetic Nervous System (PNS) activity. High vagal tone is associated with better emotion regulation, attention, and executive function.[5] Breathing practices that emphasize slow, deep breathing enhance RSA and vagal tone, thereby promoting cognitive and emotional regulation.

Neuroplasticity refers to the brain's ability to reorganize itself by forming new neural connections. Neurogenesis is the birth of new neurons that occurs in specific brain regions such as the hippocampus, which is crucial for memory and learning.[6] Evidence suggests that breathing practices enhance neuroplasticity and neurogenesis, possibly through increased brain-derived neurotrophic factor (BDNF) levels.[7] These changes underlie improvements in cognitive functions observed with regular practice. Functional magnetic resonance imaging (fMRI) and electroencephalography (EEG) studies provide insights into the neural correlates of breathing practices. These modalities have shown that controlled breathing activates the prefrontal cortex involved in attention, emotion regulation, and executive function.[8] Increased activity in the PFC and decreased amygdala activation have been observed, indicating enhanced cognitive control and reduced emotional reactivity.

Numerous clinical trials and experimental studies have investigated the effects of various breathing practices on cognitive functions. These studies encompass diverse populations, including healthy individuals, elderly adults, and patients with cognitive impairments or psychiatric conditions. The findings consistently indicate that regular practice of controlled breathing enhances attention, memory, executive function, and emotional regulation.[8-10] Moreover, the cognitive benefits of breathing practices are often accompanied by improvements in mood, stress resilience, and overall quality of life.

This review aimed to provide a comprehensive overview of the neurobiological mechanisms by which breathing practices influence cognitive functions. It critically evaluated the ancient literature, current evidence from clinical trials, neuroimaging studies, and experimental research. Additionally, the review discussed the potential applications of breathing practices in clinical settings for the prevention and treatment of cognitive decline and related disorders. By elucidating the neurobiology of breathing practices, this review tried to bridge the gap between ancient wisdom and modern neuroscience, offering evidence-based insights into the cognitive benefits of controlled breathing.

Historical Perspective on Breathing Practices

Breathing practices have a rich historical tapestry, with roots in ancient civilizations. Eastern philosophies, particularly those of India and China, integrated breath control (pranayama and qigong, respectively) as a cornerstone of physical, mental, and spiritual well-being. These practices were believed to influence vital energy, consciousness, and health. While Western civilizations, acknowledging the importance of respiration for physiological functions, have traditionally focused less on conscious breath control. However, recent centuries have witnessed a resurgence of interest, with the emergence of techniques like breathwork and the integration of breathing exercises into therapeutic modalities. This contemporary revival is underpinned by growing scientific evidence supporting the link between breath and physiological, psychological, and emotional states. Overall, the history of breathing practices reflects a global recognition of the breath as a vital life force, with its role evolving from spiritual and philosophical concepts to contemporary scientific exploration.

Pranayama: Ancient India Breathing Practice

The concept of pranayama, central to the Indian yogic tradition, originates from the Vedic texts, dating back to around 1500 BCE. The Vedas, comprising the Rigveda,

Yajurveda, Samaveda, and Atharvaveda, are some of the oldest sacred scriptures in the world. They mention the regulation of breath as a vital aspect of spiritual practices aimed at controlling prana, the life force. During the Vedic period, breath control was linked to rituals and spiritual practices. The Upanishads, philosophical texts that expanded on the Vedic ideas, further elaborated on pranayama. The Chandogya Upanishad and the Brihadaranyaka Upanishad, in particular, highlight the significance of breath control for achieving higher states of consciousness and realizing the self.[11,12] Bhagavad Gita, a sacred Hindu scripture, provides a more structured approach to pranayama. In the sixth chapter, Lord Krishna discusses the importance of breath control in meditation and spiritual discipline. This text, dating from around 500 BCE, emphasizes pranayama as essential for mental clarity and spiritual growth. It emphasizes that pranayama has the potential to regulate the mind and emotions and hence leads to an equilibrium state of the mind.[13] The equilibrium state of mind promotes cognitive function by facilitating a balance between cognitive processes and emotional responses, essential for effective decision-making. This state of mental balance optimizes neural activity by aligning existing cognitive frameworks with new information, encouraging adaptive learning through both assimilation and accommodation. Moreover, maintaining equilibrium reduces stress and anxiety, which can interfere with focus and memory recall, thereby enhancing overall cognitive efficiency.[14] Consequently, achieving mental equilibrium fosters resilience and adaptability, allowing individuals to navigate complex mental tasks more effectively and improving overall cognitive performance.

Mahrashi Patanjali compiled Yoga Sutras in the 2nd century BCE and systematically outlined the practice of yoga, with pranayama being one of the eight limbs (Ashtanga Yoga). Patanjali describes pranayama as the bridge between physical practices (asanas) and the higher states of meditation (dhyana) and enlightenment (samadhi). The breath, termed "Prana," signifies vital energy representing the conscious essence that infuses the entire universe. The literal meaning of Pranayama is breath control or expansion of breath comprises techniques focused on consciously regulating various aspects of breathing, including its frequency, depth, and the ratio between inhalation and exhalation. Yoga sutras highlight various techniques for regulating the breath, each serving to calm the mind and prepare it for deeper meditative states.[15] The Hatha Yoga Pradipika, written by Swami Swatmarama in the 15th century, is a seminal text that provides detailed instructions on various pranayama techniques. It describes methods such as Nadi Shodhana (alternate nostril breathing), Kapalabhati (skull shining breath), and Bhastrika (bellows breath), emphasizing their physical and mental benefits. According to the text, Pranayama purifies the nadis (energy channels), ensuring a free flow of prana, which stabilizes and balances mental functions. It suggests that prana directly influences the mind. By regulating the breath, one can control prana, which in turn stabilizes and balances mental functions. The verse "Chale vate chalam chittam, nischale nischalam bhavet" emphasizes the connection between breath (vata) and the mind (chitta). It translates to: "When the breath is unsteady, the mind is unsteady. When the breath is steady, the mind is steady." This verse highlights the profound relationship between breath control and mental stability. By mastering pranayama (breath control), one can achieve a calm and focused mind, which is essential for deeper meditation and spiritual growth. The practice of regulating breath is thus a gateway to controlling the mind and achieving inner peace.[16] This regulation is crucial for maintaining mental calmness and cognitive stability.

In the modern era, yoga has gained global popularity, with pranayama being an integral component of many yoga practices. Modern yoga pioneers like Swami Vivekananda, Tirumalai Krishnamacharya, and B.K.S. Iyengar have played crucial roles in adapting and popularizing pranayama techniques worldwide, highlighting their therapeutic potential approach for stress reduction, mental clarity, and overall well-being.

Traditional Chinese Practices: Qigong and Tai Chi

Breathing practices are also deeply rooted in Chinese culture, particularly within the traditions of Qigong and Tai Chi. These practices emphasize the cultivation and regulation of qi (life energy) through breath control, movement, and meditation. The origins of Qigong can be traced back over 4,000 years, with references found in ancient Chinese texts such as the Dao De Jing (Tao Te Ching) by Laozi (circa 6th century BCE) and the Huangdi Neijing (Yellow Emperor's Classic of Internal Medicine) from the 2nd century BCE.[17,18] These texts discuss the importance of harmonizing breath to balance internal energy and maintain health. The Dao De Jing, in particular,

emphasizes the concept of "wu wei" (effortless action) and the natural flow of breath as fundamental to achieving harmony with the Dao (the Way). Qigong, which means "energy work" or "energy cultivation," involves various techniques of breath control, meditation, and physical postures to enhance the flow of qi. The practice has evolved into different styles, each with specific objectives, such as medical Qigong for healing, martial Qigong for strength, and spiritual Qigong for enlightenment. Breathing techniques in Qigong are designed to open the meridians (energy pathways) and balance the body's energy system, promoting physical health and mental clarity.[19] Tai Chi, developed in the 16th century, is a martial art that incorporates slow, deliberate movements, coordinated with deep breathing and mindfulness. Originally practiced for self-defense, Tai Chi has evolved into a popular form of exercise known for its health benefits. The breath in Tai Chi is synchronized with movement to enhance energy flow, improve circulation, and calm the mind. This coordination of breath and movement is believed to balance the yin and yang energies, promoting overall harmony and well-being.[20]

Ancient Greece and Western Traditions

Breathing practices were also recognized in ancient Greece for their health benefits. Greek philosophers and physicians explored the relationship between breath and health, laying the groundwork for later Western scientific explorations. Philosophers such as Pythagoras (circa 570-495 BCE) and Plato (circa 427-347 BCE) recognized the importance of breath control in achieving mental clarity and physical health. Pythagoras, known for his contributions to mathematics and philosophy, believed that controlled breathing could harmonize the body and mind. Plato, in his dialogues, discussed the role of breath in maintaining balance and health. Hippocrates (circa 460-370 BCE), often regarded as the father of modern medicine, emphasized the importance of breath and its regulation for health. He believed that proper breathing was essential for balancing the bodily humors and maintaining overall well-being. This perspective was further developed by Galen (circa 129-200 CE), a prominent Roman physician who integrated breath control into his therapeutic practices, recognizing its benefits for both physical and mental health.[21]

Medieval Europe and the Renaissance

Breathing practices continued to be valued in medieval Europe and the Renaissance period. The revival of interest in classical Greek and Roman texts during the Renaissance led to renewed exploration of breath control and its health benefits. Medieval monastic traditions, particularly within Christianity, incorporated breathing practices into their spiritual exercises. Monks and mystics used breath control as part of their contemplative practices, aiming to achieve states of spiritual ecstasy and communion with the divine. The practice of "hesychasm" in Eastern Orthodox Christianity involved repetitive prayer combined with controlled breathing to attain inner stillness and spiritual enlightenment. The Renaissance period (14th to 17th centuries) saw a resurgence of interest in the classical traditions of Greece and Rome, including their approaches to health and breath control. Scholars such as Leonardo da Vinci (1452-1519) studied the mechanics of breathing, linking it to physical and mental health. Leonardo's anatomical drawings and observations on respiration contributed to a deeper understanding of the physiological aspects of breath control. The 19th and 20th centuries marked a significant shift in the understanding and practice of breathing techniques, driven by scientific exploration and integration into various therapeutic modalities. Modern breathing techniques emerged in the 20th century, influenced by both Eastern practices and Western scientific advancements. One notable figure is Dr. Wilhelm Reich (1897-1957), an Austrian psychoanalyst who explored the role of breath in emotional and psychological health. Reich developed "vegetotherapy," a form of therapy that used deep breathing and physical exercises to release emotional blockages and promote psychological healing.[22]

Another influential figure is Leonard Orr, who developed Rebirthing Breathwork in the 1970s. Orr's method involves conscious connected breathing to access and release suppressed emotions and traumas. Rebirthing Breathwork gained popularity as a tool for personal growth and emotional healing. Breathing practices have been integrated into various forms of psychotherapy, including Mindfulness-Based Stress Reduction (MBSR) and Mindfulness-Based Cognitive Therapy (MBCT). Developed by Jon Kabat-Zinn in the 1970s, MBSR combines mindfulness meditation with controlled breathing to reduce stress and improve mental health. MBCT developed later, integrates mindfulness practices with cognitive-behavioral techniques to prevent relapse in depression.[23]

The empirical perspective of Breathing practices

The scientific exploration of pranayama and other yoga breathing techniques has expanded significantly in recent decades. Research has demonstrated the physiological and psychological benefits of pranayama, including reduced stress[24], improved cardiovascular health[25], enhanced cognitive function[25], and better emotional regulation[26]. Studies using neuroimaging techniques such as fMRI and EEG have shown that pranayama can modulate brain activity, particularly in regions associated with emotional regulation and cognitive control[26]. Breathing techniques have been incorporated into modern medical practices for managing chronic diseases, reducing anxiety, and improving overall health. Techniques such as diaphragmatic breathing, Buteyko breathing, and respiratory muscle training are used to treat conditions like asthma[27], chronic obstructive pulmonary disease (COPD)[28], and hypertension[29]. These practices help improve lung function, enhance oxygenation, and reduce the physiological effects of stress.

Breathing practices have gained significant attention for their potential to enhance cognitive functions. These practices, which include techniques like deep breathing, pranayama, and mindfulness breathing, can positively impact brain health and cognitive performance through various neuro-psychophysiological mechanisms. Enhanced oxygenation plays a crucial role in maintaining and improving brain functions. The brain, despite constituting only about 2% of the body's weight, consumes approximately 20% of the body's oxygen supply. This high demand for oxygen is due to the brain's intensive metabolic activities, which are essential for processes such as synaptic transmission, plasticity, and overall cognitive function.[30] One of the primary ways enhanced oxygenation benefits the brain is through improved neurovascular coupling. Neurovascular coupling is a critical mechanism that ensures the brain receives adequate oxygen and nutrients to meet its metabolic demands. This process involves the coordination between neuronal activity and blood flow, where active neurons signal blood vessels to dilate, increasing local cerebral blood flow and oxygen delivery. Recent studies have highlighted the importance of neurovascular coupling in maintaining brain health. When neurons are active, they release vasoactive molecules such as nitric oxide and arachidonic acid metabolites. These molecules act on the smooth muscle cells of arterioles and capillaries, causing them to relax and dilate, thereby increasing blood flow to the active regions of the brain.[31] This increased blood flow enhances the delivery of oxygen and glucose, which are essential for neuronal function and survival.[32]

Physiological mechanisms state that the relationship between breathing and heart rate is a well-documented phenomenon known as respiratory sinus arrhythmia (RSA). RSA is characterized by the synchronization of heart rate with the breathing cycle, where heart rate increases during inhalation and decreases during exhalation. This phenomenon is mediated by the vagus nerve, a key component of the parasympathetic nervous system. High vagal tone, as indicated by greater RSA, is associated with better regulation of the cardiovascular system, reduced stress response, and improved emotional and cognitive functioning. Breathing practices, particularly those involving slow, deep, and rhythmic breathing, enhance RSA and vagal tone. This enhancement supports cognitive functions by promoting autonomic balance, reducing stress, and improving attention and executive control.

These mechanisms ensure that active regions of the brain receive an adequate supply of oxygenated blood, thereby supporting neuronal activity and metabolic needs. Enhanced oxygenation can lead to better cognitive performance, as it supports the energy demands of neurons, particularly during tasks that require high levels of concentration and mental effort.[33] Recent studies have shown that breathing practices are a potent enhancer of brain oxygenation. Exercise increases heart rate and blood flow, delivering more oxygen to the brain. This process not only supports immediate cognitive functions but also promotes long-term brain health by inducing neuroplasticity, which is the brain's ability to reorganize itself by forming new neural connections. This is particularly important in aging populations, where enhanced oxygenation through regular breathing practices can mitigate cognitive decline and reduce the risk of neurodegenerative diseases. Moreover, advanced neuroimaging techniques including functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG), have provided deeper insights into how oxygen is distributed within the brain. These techniques have revealed that sedentary lifestyles can lead to "hypoxic pockets," or areas of temporary oxygen deprivation, which are linked to an increased risk of conditions like Alzheimer's disease. Understanding these patterns can help in developing targeted interventions to improve brain oxygenation and overall cognitive health.

Therefore, enhanced oxygenation is vital for optimal brain function. It supports neuronal activity, promotes neuroplasticity, and can potentially protect against cognitive decline. Regular breathing practices and advanced imaging techniques are key tools in understanding and enhancing brain oxygenation.

The oxygen-rich brain helps an individual enhance neuroplasticity and neurogenesis through various mechanisms, including the modulation of neuro-transmitters, neurotrophic factors like brain-derived neurotrophic factor (BDNF), and stress reduction. Increased BDNF levels, in particular, are associated with improved cognitive functions and resilience to neurodegenerative processes. Neuroimaging studies provide valuable insights into the neural correlates of breathing practices. Techniques such as fMRI and EEG have shown that controlled breathing activates brain regions involved in cognitive control, emotion regulation, and attention. The fMRI studies have demonstrated increased activity in the prefrontal cortex (PFC) and decreased activity in the amygdala during and after controlled breathing exercises. The PFC is crucial for executive functions, including decision-making, problem-solving, and impulse control, while the amygdala is involved in the processing of emotions, particularly fear and stress. These findings suggest that breathing practices enhance cognitive control and emotional regulation by modulating these brain regions. A growing body of clinical and experimental research supports the cognitive benefits of breathing practices. Studies have shown that regular practice of controlled breathing can improve various aspects of cognitive function, including attention, memory, and executive functions.

Breathing practices particularly those rooted in traditional disciplines like yoga, have been increasingly recognized for their broad benefits on mental and emotional well-being. In healthy individuals, these practices are linked not only to enhanced cognitive performance but also to reductions in anxiety and improvements in mood.[34] The mechanisms behind these effects include the regulation of the autonomic nervous system, increased oxygenation of the brain[35], and the promotion of mindfulness and relaxation[36].

In clinical populations, including those with anxiety disorders, depression, and various cognitive impairments, breathing practices have shown considerable promise. Studies indicate that regular practice can alleviate symptoms, improve cognitive function[37], and contribute to overall emotional stability. These findings are significant given the growing interest in non-pharmacological interventions for mental health. Furthermore, in aging populations, breathing practices are being explored as a potential tool to mitigate age-related cognitive decline. Research suggests that they can support healthy aging by enhancing cognitive resilience, reducing stress, and promoting neuroplasticity.[38,39] As the global population ages, such interventions may become increasingly vital in maintaining cognitive health and quality of life in older adults.

CONCLUSION

Breathing exercises are a promising and easily accessible approach for enhancing cognitive functions, primarily through their substantial effects on brain neurobiology. These practices influence cognitive performance by modulating the autonomic nervous system and the hypothalamic-pituitary-adrenal (HPA) axis, while also fostering neuroplasticity in crucial brain areas. Research indicates that conscious breathing affects vital neural structures, such as the prefrontal cortex and hippocampus, which are essential for executive functioning, memory consolidation, and attention control.

The modulation of brain activity via these techniques is closely linked to the regulation of respiratory rhythms, which align with neural oscillations. This inter-connectedness of respiration and cognitive function underscores the complex relationship between physiological processes and cognitive outcomes. Breathing practices may also provide a practical intervention to counteract cognitive decline in both clinical and aging populations. Thus, further investigation utilizing neuroimaging methodologies like EEG, fNIRS, fMRI, PET, and TCD is warranted to explore these effects on executive function.

CONFLICT OF INTEREST

None

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